

# Sritiz Sahu

Full-Stack Developer / ML Researcher

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## Professional Summary

Performance-driven Full-Stack Developer and Machine Learning Researcher with a Minor in Computer Science. Specialized in Computer Vision (CV), Medical Imaging, and Deep Learning research. Proven expertise in building production-ready MERN applications for 3,000+ users and benchmarking SOTA architectures (UNet, ResNet, VGG). Skilled in developing explainable AI (XAI) systems and investigating latent space architectures for generative modeling.

## Technical Skills

|                 |                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------|
| AI & ML         | Deep Learning, Computer Vision, CNNs (ResNet, VGG16), UNet, VAEs, Explainable AI (SHAP) |
| Full-Stack      | React.js, Node.js, Express.js, Redux Toolkit, FastAPI, Tailwind CSS, ShadCN UI          |
| Languages       | Python (NumPy, Pandas, Scikit-learn), Core C++ (DSA), SQL, JavaScript (ES6+)            |
| Infrastructure  | MongoDB GridFS, JWT Authentication, RBAC, RESTful API Design, Swagger, Git/GitHub       |
| CS Fundamentals | System Design, Operating Systems, DBMS Indexing, Computer Networks, OOP                 |

## Research & Experience

Dec 2025 – **Research Intern, IIT Patna, Patna**

Jan 2026 Supervisor: Prof. M. H. Kolekar

- **Top Highlight:** Benchmarked 10+ State-of-the-Art (SOTA) Computer Vision architectures including **Attention UNet** and **MDPNet**; identified structural bottlenecks that led to a **12% gain in segmentation accuracy**.
- Synthesized 15+ publication-standard technical diagrams for a book chapter on **Generative AI and VAEs**, focusing on **latent space manifold learning** for high-security surveillance.
- Performed rigorous literature synthesis on multi-modal medical image segmentation and anomaly detection, contributing to the development of a novel research framework.
- Investigated encoder-decoder bottlenecks in medical imaging datasets to optimize gradient flow during backpropagation.

Aug 2025 – **Backend Developer, Xceed (Official NITJ Web Dev Club), Jalandhar**

- Present
- Scaled high-traffic registration platforms serving **3,000+ students**, maintaining 99.9% system uptime during peak load semester registration windows.
  - Refactored monolithic backend into modular controllers using Express.js, decreasing technical debt and boosting maintenance efficiency by 25%.
  - Deployed the VISTA ECE portal with responsive design principles, increasing accessibility for 50+ faculty members via cloud-native asset management.

Aug 2025 – **Full-Stack Intern, NIT Jalandhar TPO Website Team, Jalandhar**

- Present
- Engineered an admin-facing Dashboard to automate recruitment for 100+ firms, reducing workload by 40%.
  - Optimized MongoDB aggregation pipelines, achieving an **80% reduction in query latency** (under 200ms).
  - Architected secure middleware using Bcrypt and JWT for 100% data isolation of university records.

## Machine Learning & Full-Stack Projects

- Jan 2026 **SmartDocs: AI-Powered File Orchestrator**, *MERN Stack, OCR, MongoDB GridFS*
- Orchestrated a scalable storage engine using **MongoDB GridFS** to handle 500+ binary PDF assets without server-side memory overflow.
  - Implemented an asynchronous OCR pipeline using Tesseract.js, achieving **94% classification accuracy** through custom heuristic data normalization.
- Oct 2025 **Brain Tumor Detection via Transfer Learning**, *Python, TensorFlow, Keras*
- Leveraged the **VGG16 architecture** via Transfer Learning to classify MRI scans, implementing a custom classification head with Dropout layers to mitigate overfitting.
  - Engineered a custom Python data generator for real-time preprocessing and augmentation, ensuring efficient memory management for large-scale medical datasets.
- Aug 2025 **Advanced Object Classifier (CIFAR-10)**, *Python, ResNet50, NumPy*
- Developed a robust classification system by fine-tuning a **ResNet50 backbone**, achieving a high **97.2% test accuracy**.
  - Architected a data pipeline for 50,000+ images, utilizing NumPy for vectorized normalization and multi-stage image preprocessing.
- Feb 2026 **Trust Guard: Explainable AI Portal**, *FastAPI, SHAP, Scikit-learn*
- Developed a production-grade anomaly detection portal, integrating **Explainable AI (XAI)** using SHAP visualizations to provide model transparency for 100% of automated decisions.

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## Education

2024 – 2028 **B.Tech in Instrumentation & Control Engineering**, *NIT Jalandhar*, GPA: 8.0/10.0  
Minor in Computer Science

Relevant Coursework    Neural Networks, Deep Learning, Data Structures, Operating Systems, DBMS

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## Achievements

- **Project Leadership:** Successfully led the TPO Faculty Dashboard deployment within a strict 14-day window.
- **Technical Contributions:** Active contributor to campus open-source initiatives serving 3,000+ users.
- **Competitive Programming:** Proficient in C++ algorithmic optimization for time-critical applications.